

3.3 *Inverse of an increasing convex function.* Suppose $f : \mathbf{R} \rightarrow \mathbf{R}$ is increasing and convex on its domain (a, b) . Let g denote its inverse, *i.e.*, the function with domain $(f(a), f(b))$ and $g(f(x)) = x$ for $a < x < b$. What can you say about convexity or concavity of g ?

3.10 *An extension of Jensen's inequality.* One interpretation of Jensen's inequality is that randomization or dithering hurts, *i.e.*, raises the average value of a convex function: For f convex and v a zero mean random variable, we have $\mathbf{E} f(x_0 + v) \geq f(x_0)$. This leads to the following conjecture. If f is convex, then the larger the variance of v , the larger $\mathbf{E} f(x_0 + v)$.

- (a) Give a counterexample that shows that this conjecture is false. Find zero mean random variables v and w , with $\mathbf{var}(v) > \mathbf{var}(w)$, a convex function f , and a point x_0 , such that $\mathbf{E} f(x_0 + v) < \mathbf{E} f(x_0 + w)$.

- (b) The conjecture is true when v and w are scaled versions of each other. Show that $\mathbf{E} f(x_0 + tv)$ is monotone increasing in $t \geq 0$, when f is convex and v is zero mean.

3.16 For each of the following functions determine whether it is convex, concave, quasiconvex, or quasiconcave.

(a) $f(x) = e^x - 1$ on $\mathbf{R}$.

(b) $f(x_1, x_2) = x_1 x_2$ on $\mathbf{R}_{++}^2$.

(c) $f(x_1, x_2) = 1/(x_1 x_2)$ on $\mathbf{R}_{++}^2$.

(d) $f(x_1, x_2) = x_1/x_2$ on $\mathbf{R}_{++}^2$.

(e) $f(x_1, x_2) = x_1^2/x_2$ on $\mathbf{R} \times \mathbf{R}_{++}$.

(f) $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, where $0 \leq \alpha \leq 1$, on $\mathbf{R}_{++}^2$.

3.24 *Some functions on the probability simplex.* Let x be a real-valued random variable which takes values in $\{a_1, \dots, a_n\}$ where $a_1 < a_2 < \dots < a_n$, with $\mathbf{prob}(x = a_i) = p_i$, $i = 1, \dots, n$. For each of the following functions of p (on the probability simplex $\{p \in \mathbf{R}_+^n \mid \mathbf{1}^T p = 1\}$), determine if the function is convex, concave, quasiconvex, or quasiconcave.

- (a) $\mathbf{E} x$.
- (b) $\mathbf{prob}(x \geq \alpha)$.
- (c) $\mathbf{prob}(\alpha \leq x \leq \beta)$.
- (d) $\sum_{i=1}^n p_i \log p_i$, the negative entropy of the distribution.
- (e) $\mathbf{var} x = \mathbf{E}(x - \mathbf{E} x)^2$.
- (f) $\mathbf{quartile}(x) = \inf\{\beta \mid \mathbf{prob}(x \leq \beta) \geq 0.25\}$.
- (g) The cardinality of the smallest set $\mathcal{A} \subseteq \{a_1, \dots, a_n\}$ with probability $\geq 90\%$. (By cardinality we mean the number of elements in $\mathcal{A}$.)
- (h) The minimum width interval that contains 90% of the probability, *i.e.*,

$$\inf \{\beta - \alpha \mid \mathbf{prob}(\alpha \leq x \leq \beta) \geq 0.9\}.$$

3.32 *Products and ratios of convex functions.* In general the product or ratio of two convex functions is not convex. However, there are some results that apply to functions on $\mathbf{R}$. Prove the following.

- (a) If f and g are convex, both nondecreasing (or nonincreasing), and positive functions on an interval, then fg is convex.
- (b) If f, g are concave, positive, with one nondecreasing and the other nonincreasing, then fg is concave.
- (c) If f is convex, nondecreasing, and positive, and g is concave, nonincreasing, and positive, then f/g is convex.

4.1 Consider the optimization problem

$$\begin{array}{ll}\text{minimize} & f_0(x_1, x_2) \\ \text{subject to} & 2x_1 + x_2 \geq 1 \\ & x_1 + 3x_2 \geq 1 \\ & x_1 \geq 0, \quad x_2 \geq 0.\end{array}$$

Make a sketch of the feasible set. For each of the following objective functions, give the optimal set and the optimal value.

(a) $f_0(x_1, x_2) = x_1 + x_2.$

(b) $f_0(x_1, x_2) = -x_1 - x_2.$

(c) $f_0(x_1, x_2) = x_1.$

(d) $f_0(x_1, x_2) = \max\{x_1, x_2\}.$

(e) $f_0(x_1, x_2) = x_1^2 + 9x_2^2.$

4.3 Prove that $x^* = (1, 1/2, -1)$ is optimal for the optimization problem

$$\begin{array}{ll}\text{minimize} & (1/2)x^T Px + q^T x + r \\ \text{subject to} & -1 \leq x_i \leq 1, \quad i = 1, 2, 3,\end{array}$$

where

$$P = \begin{bmatrix} 13 & 12 & -2 \\ 12 & 17 & 6 \\ -2 & 6 & 12 \end{bmatrix}, \quad q = \begin{bmatrix} -22.0 \\ -14.5 \\ 13.0 \end{bmatrix}, \quad r = 1.$$

4.5 Equivalent convex problems. Show that the following three convex problems are equivalent. Carefully explain how the solution of each problem is obtained from the solution of the other problems. The problem data are the matrix $A \in \mathbf{R}^{m \times n}$ (with rows a_i^T), the vector $b \in \mathbf{R}^m$, and the constant $M > 0$.

(a) The *robust least-squares problem*

$$\text{minimize} \quad \sum_{i=1}^m \phi(a_i^T x - b_i),$$

with variable $x \in \mathbf{R}^n$, where $\phi : \mathbf{R} \rightarrow \mathbf{R}$ is defined as

$$\phi(u) = \begin{cases} u^2 & |u| \leq M \\ M(2|u| - M) & |u| > M. \end{cases}$$

(This function is known as the *Huber penalty function*; see §6.1.2.)

(b) The *least-squares problem with variable weights*

$$\begin{aligned} &\text{minimize} && \sum_{i=1}^m (a_i^T x - b_i)^2 / (w_i + 1) + M^2 \mathbf{1}^T w \\ &\text{subject to} && w \succeq 0, \end{aligned}$$

with variables $x \in \mathbf{R}^n$ and $w \in \mathbf{R}^m$, and domain $\mathcal{D} = \{(x, w) \in \mathbf{R}^n \times \mathbf{R}^m \mid w \succ -\mathbf{1}\}$.

Hint. Optimize over w assuming x is fixed, to establish a relation with the problem in part (a).

(This problem can be interpreted as a weighted least-squares problem in which we are allowed to adjust the weight of the i th residual. The weight is one if $w_i = 0$, and decreases if we increase w_i . The second term in the objective penalizes large values of w , *i.e.*, large adjustments of the weights.)

(c) The *quadratic program*

$$\begin{aligned} &\text{minimize} && \sum_{i=1}^m (u_i^2 + 2Mv_i) \\ &\text{subject to} && -u - v \preceq Ax - b \preceq u + v \\ &&& 0 \preceq u \preceq M\mathbf{1} \\ &&& v \succeq 0. \end{aligned}$$

4.7 *Convex-concave fractional problems.* Consider a problem of the form

$$\begin{array}{ll}\text{minimize} & f_0(x)/(c^T x + d) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

where $f_0, f_1, \dots, f_m$ are convex, and the domain of the objective function is defined as $\{x \in \mathbf{dom} f_0 \mid c^T x + d > 0\}$.

- (a) Show that this is a quasiconvex optimization problem.
- (b) Show that the problem is equivalent to

$$\begin{array}{ll}\text{minimize} & g_0(y, t) \\ \text{subject to} & g_i(y, t) \leq 0, \quad i = 1, \dots, m \\ & Ay = bt \\ & c^T y + dt = 1,\end{array}$$

where g_i is the perspective of f_i (see §3.2.6). The variables are $y \in \mathbf{R}^n$ and $t \in \mathbf{R}$. Show that this problem is convex.

- (c) Following a similar argument, derive a convex formulation for the *convex-concave* fractional problem

$$\begin{array}{ll}\text{minimize} & f_0(x)/h(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m \\ & Ax = b\end{array}$$

where $f_0, f_1, \dots, f_m$ are convex, h is concave, the domain of the objective function is defined as $\{x \in \mathbf{dom} f_0 \cap \mathbf{dom} h \mid h(x) > 0\}$ and $f_0(x) \geq 0$ everywhere.

As an example, apply your technique to the (unconstrained) problem with

$$f_0(x) = (\mathbf{tr} F(x))/m, \quad h(x) = (\det(F(x)))^{1/m},$$

with $\mathbf{dom}(f_0/h) = \{x \mid F(x) \succ 0\}$, where $F(x) = F_0 + x_1 F_1 + \dots + x_n F_n$ for given $F_i \in \mathbf{S}^m$. In this problem, we minimize the ratio of the arithmetic mean over the geometric mean of the eigenvalues of an affine matrix function $F(x)$.

5.27 *Equality constrained least-squares.* Consider the equality constrained least-squares problem

$$\begin{array}{ll}\text{minimize} & \|Ax - b\|_2^2 \\ \text{subject to} & Gx = h\end{array}$$

where $A \in \mathbf{R}^{m \times n}$ with $\mathbf{rank} A = n$, and $G \in \mathbf{R}^{p \times n}$ with $\mathbf{rank} G = p$.

Give the KKT conditions, and derive expressions for the primal solution $x^\star$ and the dual solution $\nu^\star$.

5.29 The problem

$$\begin{array}{ll}\text{minimize} & -3x_1^2 + x_2^2 + 2x_3^2 + 2(x_1 + x_2 + x_3) \\ \text{subject to} & x_1^2 + x_2^2 + x_3^2 = 1,\end{array}$$

is a special case of (5.32), so strong duality holds even though the problem is not convex. Derive the KKT conditions. Find all solutions x, ν that satisfy the KKT conditions. Which pair corresponds to the optimum?

5.30 Derive the KKT conditions for the problem

$$\begin{array}{ll} \text{minimize} & \text{tr } X - \log \det X \\ \text{subject to} & Xs = y, \end{array}$$

with variable $X \in \mathbf{S}^n$ and domain $\mathbf{S}_{++}^n$. $y \in \mathbf{R}^n$ and $s \in \mathbf{R}^n$ are given, with $s^T y = 1$. Verify that the optimal solution is given by

$$X^* = I + yy^T - \frac{1}{s^T s} ss^T.$$

5.31 *Supporting hyperplane interpretation of KKT conditions.* Consider a convex problem with no equality constraints,

$$\begin{array}{ll} \text{minimize} & f_0(x) \\ \text{subject to} & f_i(x) \leq 0, \quad i = 1, \dots, m. \end{array}$$

Assume that $x^* \in \mathbf{R}^n$ and $\lambda^* \in \mathbf{R}^m$ satisfy the KKT conditions

$$\begin{array}{rcll} f_i(x^*) & \leq & 0, & i = 1, \dots, m \\ \lambda_i^* & \geq & 0, & i = 1, \dots, m \\ \lambda_i^* f_i(x^*) & = & 0, & i = 1, \dots, m \\ \nabla f_0(x^*) + \sum_{i=1}^m \lambda_i^* \nabla f_i(x^*) & = & 0. \end{array}$$

Show that

$$\nabla f_0(x^*)^T (x - x^*) \geq 0$$

for all feasible x . In other words the KKT conditions imply the simple optimality criterion of §4.2.3.